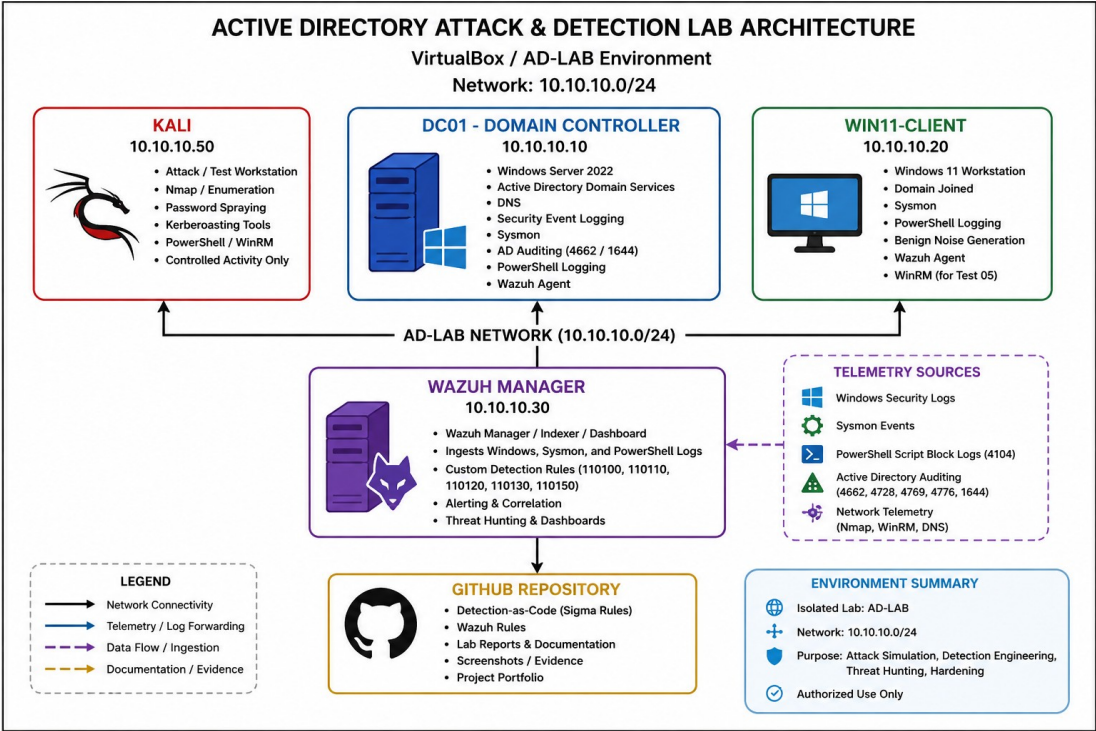


ACTIVE DIRECTORY ATTACK & DETECTION LAB

Technical Engineering Report

Detection Engineering | Active Directory Security | Wazuh SIEM | Sysmon | Sigma



High-level AD-LAB architecture and monitoring topology. Evidence: diagrams/architecture.png

Document Field	Value
Environment	AD-LAB / adlab.test
Network	10.10.10.0/24
Domain Controller	DC01 - 10.10.10.10
Domain Workstation	WIN11-CLIENT - 10.10.10.20
SIEM	WAZUH - 10.10.10.30
Controlled Test Host	KALI - 10.10.10.50
Report Date	16 August 2026
Repository	https://github.com/osama-git05/active-directory-attack-detection-lab

Purpose

Provide a defensible, evidence-based record of the lab architecture, telemetry pipeline, detection logic, controlled security scenarios, hunting workflows, false-positive validation, hardening actions, documented gaps, and engineering recommendations.

SECTION 0

Report Map

How this document is organized

Section	Topic	Engineering focus
1	Executive Summary	Outcome, validated detections, gaps, and final assessment
2	Authorization and Scope	Isolation, boundaries, assumptions, exclusions
3	Architecture and Identity Foundation	Hosts, network, domain, identities, SPN
4	Telemetry Engineering	Advanced Audit Policy, PowerShell, Sysmon, Wazuh ingestion
5	Detection Engineering Baseline	Custom Wazuh rules, Sigma rules, ATT&CK mapping
6	Network Reconnaissance	Nmap / Sysmon Event 3 / Wazuh correlation
7	Password Spraying	Authentication failures / frequency correlation
8	Kerberoasting	4769 / RC4 0x17 / svc_web
9	PowerShell Detection	4104 / Sigma AD recon / Wazuh rule drift
10	Remote Administrative Logon	4624 Type 3 telemetry gap
11	Privileged Group Change	4728 / Wazuh 60159 / cleanup
12	Canary Deception	4776 / custom rule 110150 / post-hardening retest
13	Targeted AD Object Auditing	4662 / SACL dependency / Wazuh visibility
14	LDAP Diagnostic Visibility	1644 / NTDS Field Engineering / ingestion gap
15	Threat Hunting	Repeatable analyst queries and investigative fields
16	Benign Noise and FP Validation	Background activity and two-rule retest
17	Hardening and Retest	Remediation actions and verification
18	Validation Matrix and Findings	Detection status, engineering observations
19	Limitations and Recommendations	Current technical debt and roadmap
20	Appendices	Rules, configs, repository artifacts, evidence catalog

SECTION 1

Executive Summary

Technical outcome and engineering conclusion

The Active Directory Attack & Detection Lab is an isolated detection-engineering environment built to show how identity, authentication, Kerberos, PowerShell, endpoint, and directory-service activity is generated, collected, correlated, investigated, tuned, and retested. The lab intentionally evaluates both successful detections and monitoring failures so the final result reflects real engineering maturity rather than a collection of screenshots.

The environment combines Windows Server 2022, Windows 11, Active Directory Domain Services, Sysmon, PowerShell Script Block Logging, Windows Advanced Audit Policy, Wazuh SIEM, custom Wazuh XML rules, Sigma rules, and a Kali Linux test host. Controlled activity was executed only against the isolated AD-LAB network.

Validation ID	Scenario	Final state
ADLAB-001	Failed authentication / password spray	Detected - custom Wazuh 110110
ADLAB-002	Kerberoasting	Detected - custom Wazuh 110100
ADLAB-003	PowerShell scenario	Detected in project validation; current Wazuh/Sigma logic needs alignment
ADLAB-004	Privileged group membership change	Detected - Wazuh built-in 60159, Level 12
ADLAB-005	Canary authentication	Detected - custom Wazuh 110150, Level 12
ADLAB-006	Remote administrative logon	Logged but not alerted
ADLAB-007	Active Directory object access	Telemetry validated / searchable in Wazuh
ADLAB-008	LDAP diagnostic query	Local telemetry validated; Wazuh ingestion not confirmed

Engineering conclusion

The strongest outcome is the complete detection-engineering lifecycle: build -> telemetry -> controlled activity -> SIEM ingestion -> detection -> hunting -> false-positive validation -> hardening -> retest -> documentation. Two visibility gaps remain explicit: remote Type-3 administrative logon alerting and LDAP Event 1644 Wazuh ingestion.

1.1 Core strengths

- Correlation-based network reconnaissance detection rather than single-connection alerting.
- Frequency-based password-spray detection across different target identities.
- Kerberoasting detection using Event 4769, RC4 encryption type 0x17, and non-machine account filtering.
- High-signal canary authentication detection using a disabled, non-privileged identity.
- Targeted Active Directory object auditing with Event 4662 and object-level SACL configuration.
- Benign-noise testing to validate detections outside a completely silent lab.
- Post-hardening detection retest demonstrating that remediation did not remove visibility.
- Transparent documentation of telemetry gaps and rule-content drift.

SECTION 2

Authorization and Scope

Security boundary, assumptions, and exclusions

All testing documented in this report was performed in an isolated and authorized virtual lab. The AD test network was designed to prevent uncontrolled interaction with production systems or public infrastructure.

Scope element	Definition
Authorized network	AD-LAB internal VirtualBox network
Subnet	10.10.10.0/24
Domain	adlab.test
Identity scope	Lab-only users, service accounts, and canary account
Attack source	Kali Linux and controlled Windows test actions
Collection destination	Wazuh manager / indexer / dashboard
Purpose	Defensive detection engineering, telemetry validation, hunting, tuning, and hardening

2.1 Explicit exclusions

- No production Active Directory systems or real organizational credentials.
- No public targets or Internet-facing attack activity.
- No LSASS credential dumping, DCSync, Golden Ticket, Silver Ticket, ransomware, or destructive payloads.
- No persistence or evasion against real environments.
- No claim that a small local lab reproduces the full background noise, scale, or identity complexity of an enterprise.

Interpretation rule

A Windows event is not automatically counted as a successful SIEM detection. The report distinguishes endpoint telemetry, central ingestion, rule matches, and dedicated alerts.

SECTION 3

Architecture and Identity Foundation

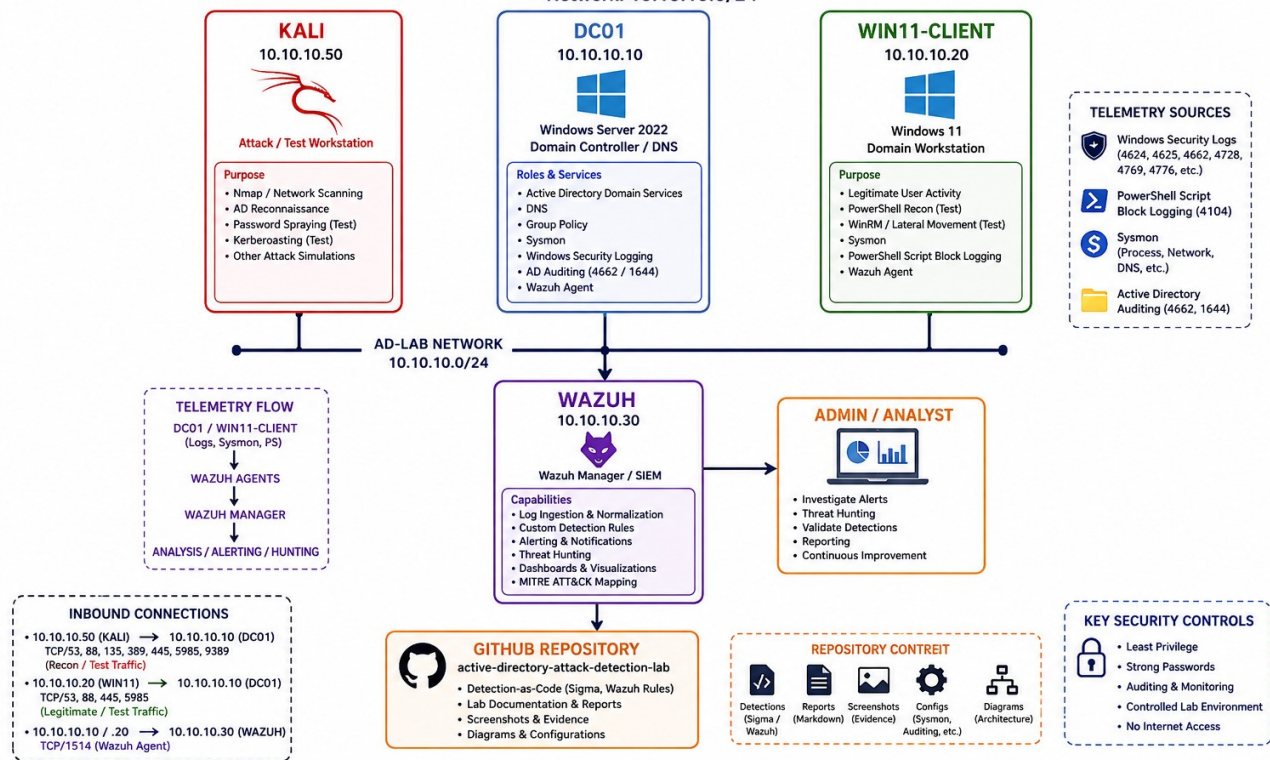
Virtualized AD environment and test identities

Host	Operating system / platform	Role	IP
DC01	Windows Server 2022	Domain Controller, DNS, Group Policy, Windows telemetry	10.10.10.10
WIN11-CLIENT	Windows 11 Pro	Domain workstation, Sysmon, PowerShell logging, Wazuh agent	10.10.10.20
WAZUH	Official Wazuh appliance	Manager, Indexer, Dashboard	10.10.10.30
KALI	Kali Linux	Controlled test workstation	10.10.10.50

ACTIVE DIRECTORY ATTACK & DETECTION LAB

AD-LAB ENVIRONMENT (VirtualBox)

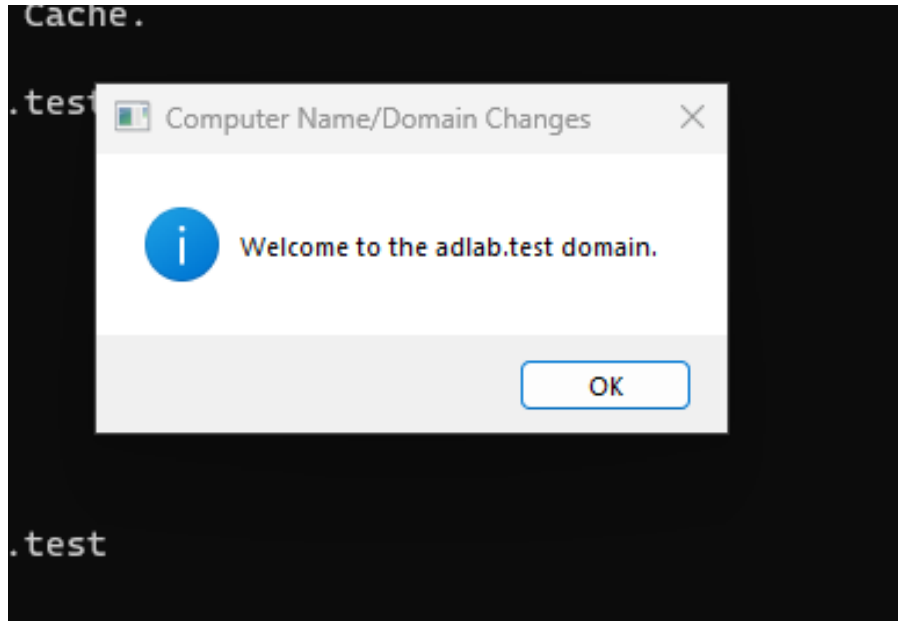
Network: 10.10.10.0/24



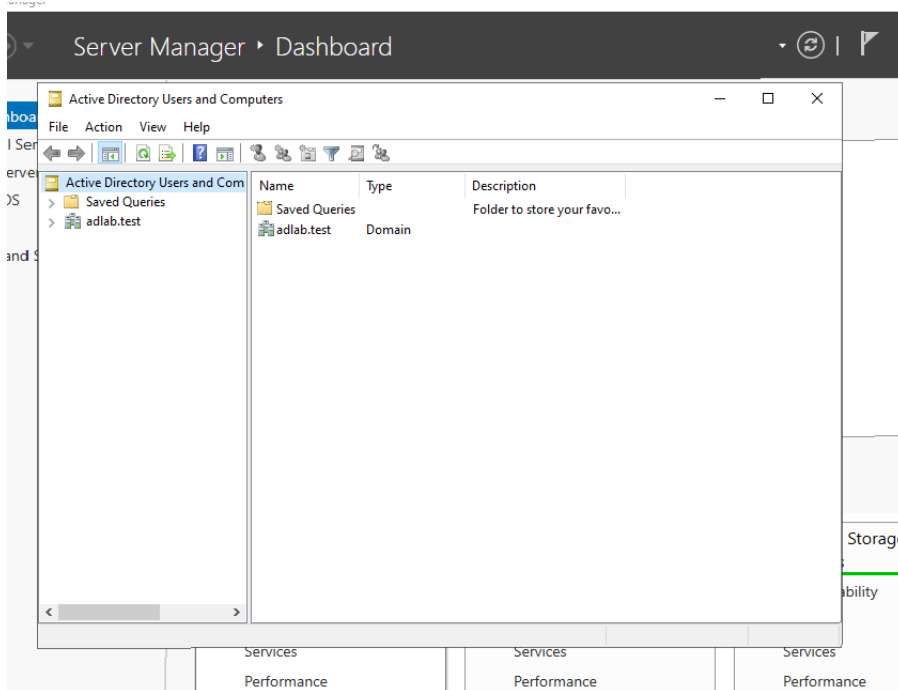
Detailed architecture showing Windows endpoints, telemetry sources, Wazuh, and the controlled test host. Evidence: diagrams/architecture-detailed.png

3.1 Domain build and workstation integration

DC01 hosts the adlab.test forest and DNS. WIN11-CLIENT was joined to the domain and used as the primary workstation for user-side telemetry, PowerShell activity, canary authentication testing, and remote administration tests.



Successful domain join confirmation for the Windows 11 workstation. Evidence: day2-domain-join.png



Windows Server management view used during Domain Controller role and service configuration. Evidence: day1-ad-ds-installed.png

```
PS C:\Users\Administrator> setspn -L ADLAB\svc_web
Registered ServicePrincipalNames for CN=svc_web,OU=Service Accounts,DC=adlab,DC=test:
HTTP/web.adlab.test
PS C:\Users\Administrator>
```

PowerShell validation of domain context and Active Directory administration state. Evidence: day1-domain-controller.png

3.2 Test identities

Identity	Purpose	Security significance	Final state
bob.user	Standard user	Normal identity activity	Enabled
helpdesk.test	Privilege-change testing	Temporary Domain Admin membership used for Event 4728 validation	Removed from Domain Admins
svc_web	Kerberos service account	SPN-bearing identity for	Enabled; non-administrative;

Identity	Purpose	Security significance	Final state
		Kerberoasting validation	password strengthened
canary.admin	Deception / tripwire	No legitimate authentication workflow	Disabled; non-privileged

HTTP/web.adlab.test
Account: ADLAB\svc_web

SECTION 4

Telemetry Engineering

Windows auditing, PowerShell, Sysmon, and Wazuh ingestion

Detection quality depends first on telemetry quality. The lab therefore establishes the event sources needed for each scenario before custom detection logic is evaluated.

Telemetry source	Primary events	Detection use
Windows Security	4624, 4625, 4662, 4672, 4688, 4728, 4732, 4756, 4768, 4769, 4771, 4776, 5136	Authentication, Kerberos, privilege changes, AD object auditing
PowerShell Operational	4104	Script Block Logging and command-content inspection
Sysmon Operational	1, 3, 11, 22	Process, network, file creation, DNS visibility
Directory Service	1644	Temporary LDAP diagnostic visibility

4.1 Advanced Audit Policy

The documented policy enables authentication, Kerberos, logon, account-management, process-creation, Directory Service Access, and Directory Service Changes auditing. Directory Service Access was validated as Success and Failure.

```
PS C:\Users\Administrator> gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

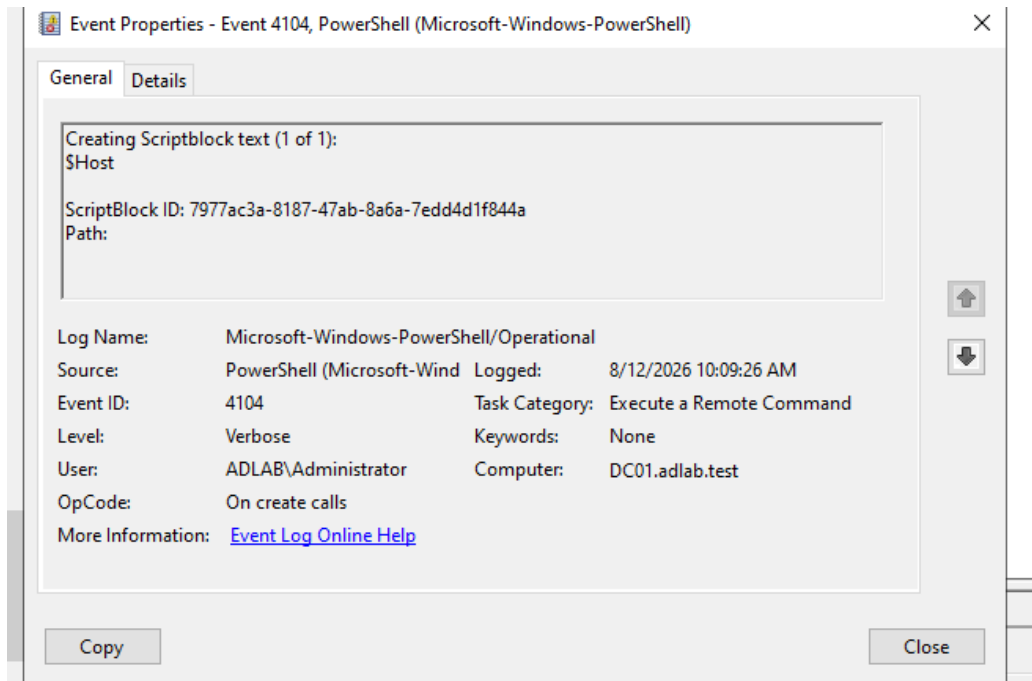
PS C:\Users\Administrator> auditpol /get /category:*
System audit policy
Category/Subcategory      Setting
System
  Security System Extension No Auditing
  System Integrity         No Auditing
  IPsec Driver             No Auditing
  Other System Events      No Auditing
  Security State Change    No Auditing
Logon/Logoff
  Logon                    Success and Failure
  Logoff                   No Auditing
  Account Lockout          No Auditing
  IPsec Main Mode          No Auditing
  IPsec Quick Mode         No Auditing
  IPsec Extended Mode      No Auditing
  Special Logon            Success
  Other Logon/Logoff Events No Auditing
  Network Policy Server    No Auditing
  User / Device Claims     No Auditing
  Group Membership         No Auditing
Object Access
  File System              No Auditing
  Registry                 No Auditing
  Kernel Object            No Auditing
  SAM                      No Auditing
  Certification Services   No Auditing
  Application Generated    No Auditing
  Handle Manipulation      No Auditing
  File Share               No Auditing
  Filtering Platform Packet Drop No Auditing
  Filtering Platform Connection No Auditing
  Other Object Access Events No Auditing
  Detailed File Share      No Auditing
  Removable Storage        No Auditing
  Central Policy Staging   No Auditing
Privilege Use
  Non Sensitive Privilege Use No Auditing
  Other Privilege Use Events No Auditing
  Sensitive Privilege Use   No Auditing
Detailed Tracking
  Process Creation         Success
```

Audit policy output used to verify Windows security subcategories required by the lab. Evidence: day3-audit-policy-1.png

```
auditpol /get /category:*
auditpol /get /subcategory:"Directory Service Access"
```

4.2 PowerShell Script Block Logging

PowerShell Script Block Logging was enabled so analysts could inspect command content rather than relying only on process names. Event ID 4104 became the primary PowerShell content source.

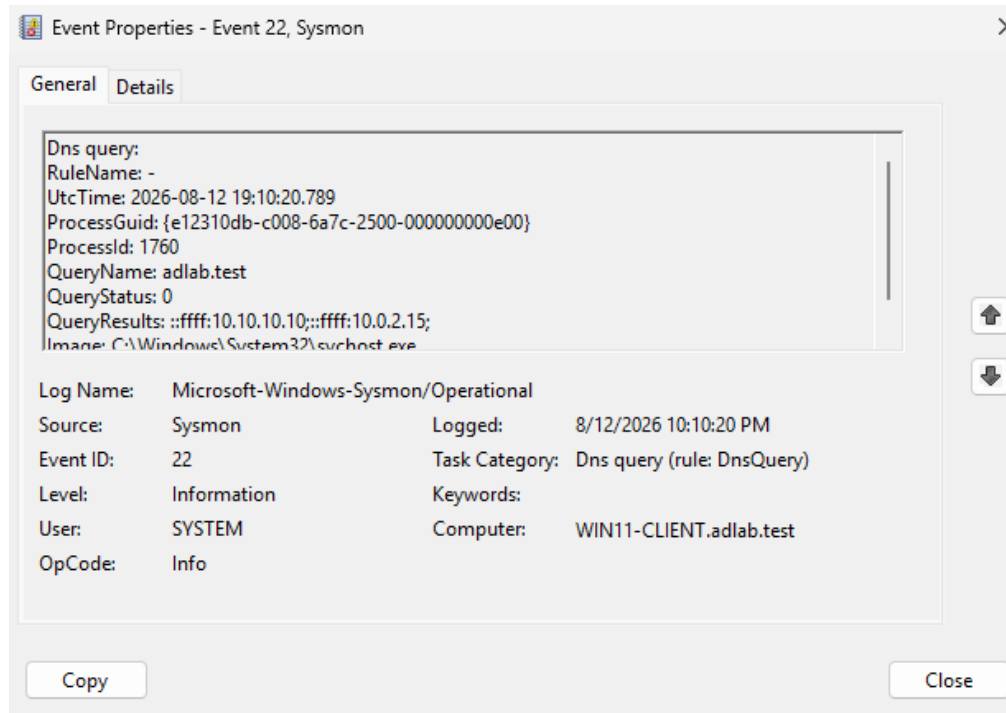


PowerShell Operational Event 4104 demonstrating script-block visibility. Evidence: day3-powershell-4104-event.png

4.3 Sysmon

The current repository configuration is intentionally broad for an isolated detection-engineering lab. It collects ProcessCreate, NetworkConnect, FileCreate, and DnsQuery telemetry and calculates SHA-256 and imphash values.

```
<Sysmon schemaversion="4.90">
  <HashAlgorithms>sha256,imphash</HashAlgorithms>
  <EventFiltering>
    <ProcessCreate onmatch="exclude" />
    <NetworkConnect onmatch="exclude" />
    <FileCreate onmatch="exclude" />
    <DnsQuery onmatch="exclude" />
  </EventFiltering>
</Sysmon>
```

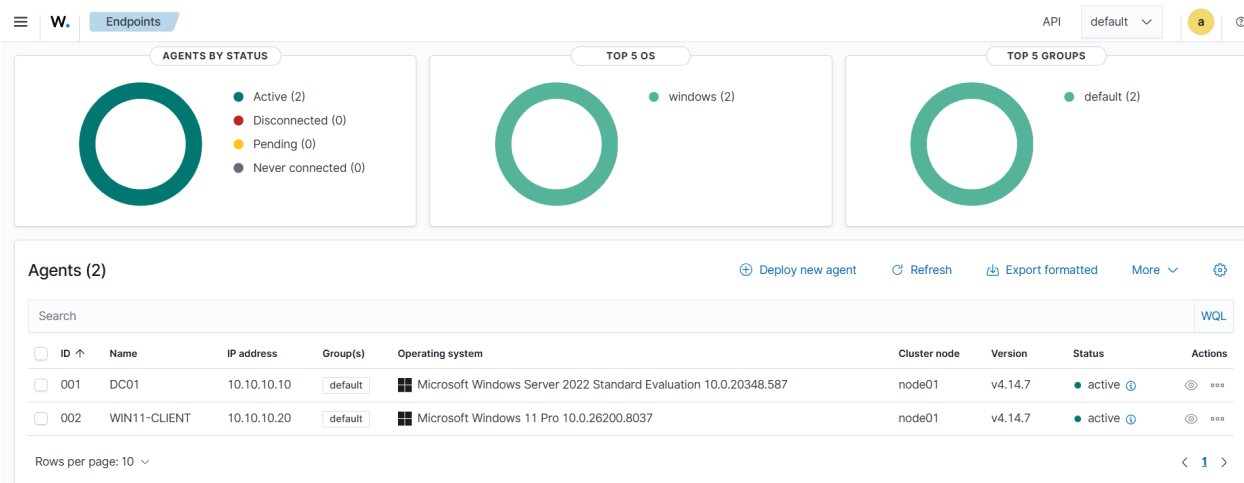


Representative Sysmon event evidence from the Windows endpoint. Evidence: day3-sysmon-win11-dns-event22.png

4.4 Wazuh agent ingestion

The repository agent configuration explicitly collects the PowerShell and Sysmon Operational channels. Windows Security telemetry is demonstrated throughout the evidence set, but the current repository agent.conf does not explicitly show the Security channel, making this a reproducibility/documentation gap.

```
<agent_config os="Windows">
  <localfile>
    <location>Microsoft-Windows-PowerShell/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>
  <localfile>
    <location>Microsoft-Windows-Sysmon/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>
</agent_config>
```



Wazuh agent status confirming monitored Windows endpoints were reporting to the SIEM. Evidence: day3-wazuh-all-agents-active.png

SECTION 5

Detection Engineering Baseline

Custom Wazuh correlation and Sigma detection-as-code

Rule / file	Purpose	Severity	ATT&CK
110100	Kerberoasting - 4769 + RC4 0x17 + non-machine account	Level 10	T1558.003
110110	Password spray correlation across different target users	Level 12	T1110.003
110120	Suspicious PowerShell script-block execution patterns	Level 10	T1059.001
110129	Hidden inbound Sysmon Event 3 helper	Level 1 / no_log	Supporting rule
110130	Network service scanning across different destination ports	Level 10	T1046
110150	Authentication against canary.admin	Level 12	T1078
kerberoasting.yml	Portable Kerberoasting detection	High	T1558.003
powershell-ad-recon.yml	Portable Get-AD* reconnaissance detection	Medium	T1087.002 / T1069.002
privileged-group-change.yml	Portable privileged-group addition detection	High	T1098

Important implementation note

The current Wazuh rule 110120 and Sigma powershell-ad-recon.yml are not semantically equivalent. The XML detects IEX / DownloadString / Net.WebClient / FromBase64String-style patterns, while Sigma detects Get-ADDomain, Get-ADUser, Get-ADGroup, and Get-ADGroupMember. This report treats that as detection-content drift requiring cleanup.

5.1 Wazuh rule validation model

The lab uses both direct event matching and correlation. The strongest examples are 110110 and 110130, which require repeated behavior within defined time windows. This reduces reliance on one-off events and demonstrates stateful SIEM logic.

5.2 Sigma role

Sigma files are included as portable detection-as-code artifacts. They were manually reviewed for syntax and logic in the project workflow; a dedicated Sigma CLI schema-validation pipeline is not claimed.

SECTION 6

Scenario 01 - Network Reconnaissance

Nmap service discovery correlated from Sysmon Event 3

Parameter	Value
Source	KALI - 10.10.10.50
Target	DC01 - 10.10.10.10
Technique	T1046 - Network Service Discovery
Telemetry	Sysmon Event 3
Detection	110129 helper -> 110130 correlation
Severity	Level 10
Result	Detected

6.1 Controlled activity

```
nmap -sV -Pn 10.10.10.10
```

```
(kali㉿kali)-[~]
$ nmap -Pn -sT -p 53,88,135,139,389,445,464,636,3268,3269 10.10.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-14 17:32 EDT
Nmap scan report for 10.10.10.10
Host is up (0.0017s latency).

PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl

Nmap done: 1 IP address (1 host up) scanned in 13.04 seconds
```

Controlled Nmap service discovery against DC01. Evidence: day4-kali-dc01-service-recon.png

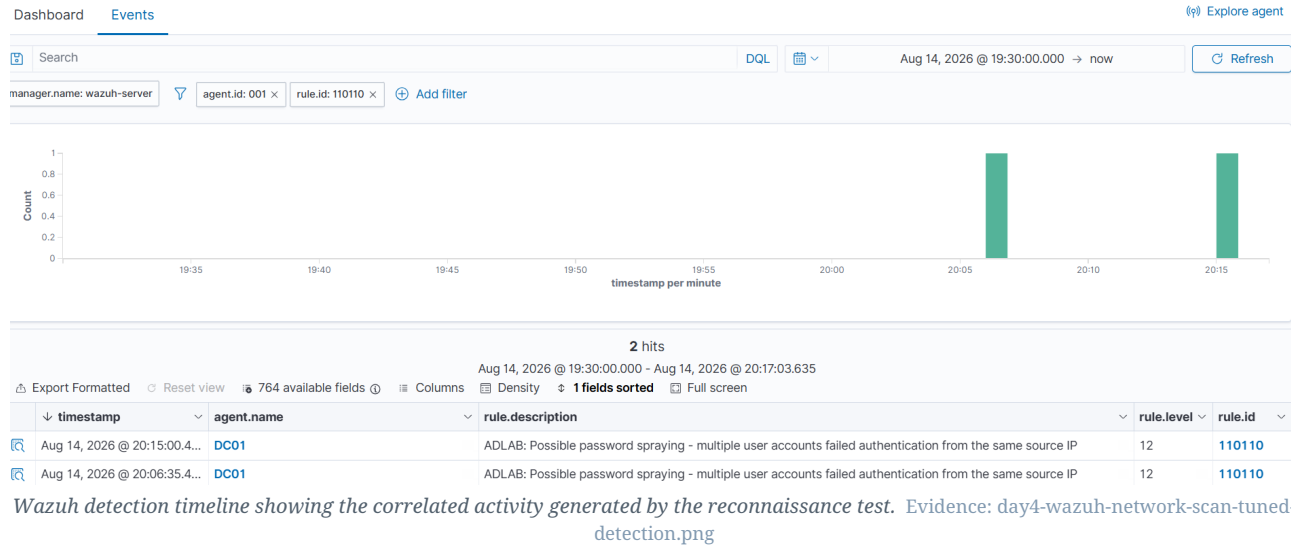
```
{
  "/var/ossec/logs/alerts/alerts.json"
{
  "timestamp": "2026-08-15T14:47:37.652+0000",
  "ruleId": "110130",
  "level": 10,
  "description": "ADLAB: Possible network service scanning from 10.10.10.50 against 10.10.10.10",
  "sourceIp": "10.10.10.50",
  "destinationIp": "10.10.10.10",
  "destinationPort": "3268"
}
```

Supporting recon telemetry showing repeated multi-port activity from the controlled source. Evidence: day4-dc01-kali-multiport-recon.png

6.2 Detection logic

```
<rule id="110129" level="1">
  <if_group>sysmon_event3</if_group>
  <field name="win.system.eventID">^3$</field>
  <field name="win.eventdata.initiated">^false$</field>
  <options>no_log</options>
</rule>
```

```
<rule id="110130" level="10" frequency="5" timeframe="10">
<if_matched_sid>110129</if_matched_sid>
<same_field>win.eventdata.sourceIp</same_field>
<same_field>win.eventdata.destinationIp</same_field>
<different_field>win.eventdata.destinationPort</different_field>
</rule>
```



Validation result: DETECTED

Rule 110130 correlated repeated inbound connections from the same source to the same destination across different ports. The rule was later tested while benign background traffic was active, with no observed benign false positive in the validation window.

SECTION 7

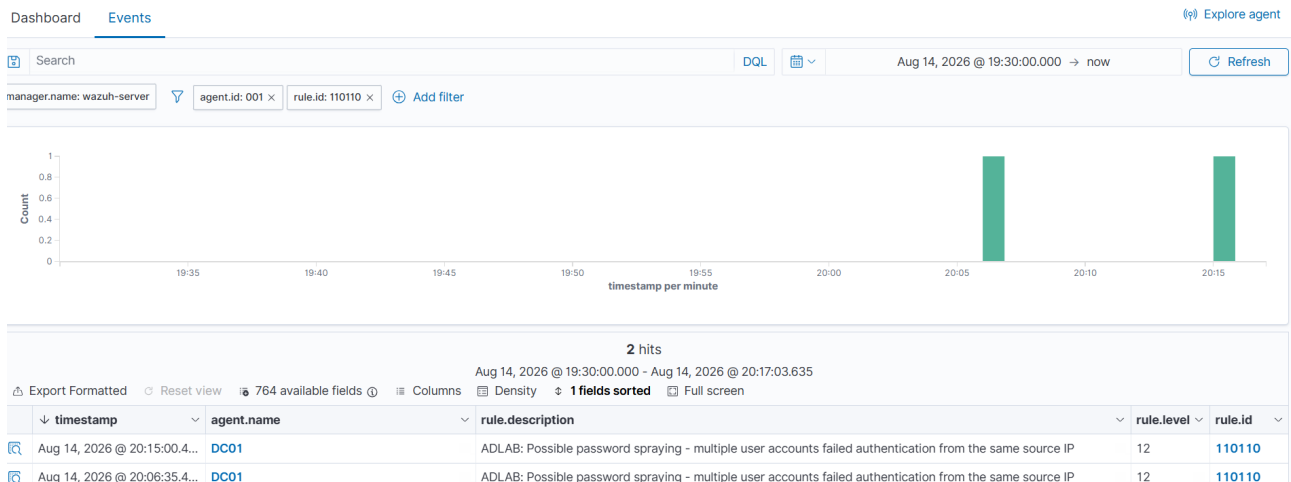
Scenario 02 - Password Spraying

Authentication-failure correlation across multiple identities

Parameter	Value
Technique	T1110.003 - Password Spraying
Relevant events	4625 / 4771 / 4776 depending on authentication path
Custom rule	110110
Correlation	4 matches in 120 seconds
Key dimensions	Same source IP; different target usernames
Severity	Level 12
Result	Detected

7.1 Detection logic

```
<rule id="110110" level="12" frequency="4" timeframe="120">
  <if_matched_sid>60122</if_matched_sid>
  <same_field>win.eventdata.ipAddress</same_field>
  <different_field>win.eventdata.targetUserName</different_field>
  <description>ADLAB: Possible password spraying - multiple user accounts failed authentication from the same source IP</description>
  <mitre><id>T1110.003</id></mitre>
</rule>
```



Wazuh dashboard evidence used during authentication-correlation validation. Evidence: day3-wazuh-password-spray-dashboard-alert.png

Tuning principle

The rule is intentionally correlation-based so isolated user mistakes do not automatically become a spray alert. Production tuning should also consider approved scanners, service accounts, lockout behavior, source-network context, and identity-management systems.

Validation result: DETECTED

The project validation matrix records custom Wazuh rule 110110 as detected for the controlled password-spray scenario.

SECTION 8

Scenario 03 - Kerberoasting

Kerberos service-ticket visibility for svc_web

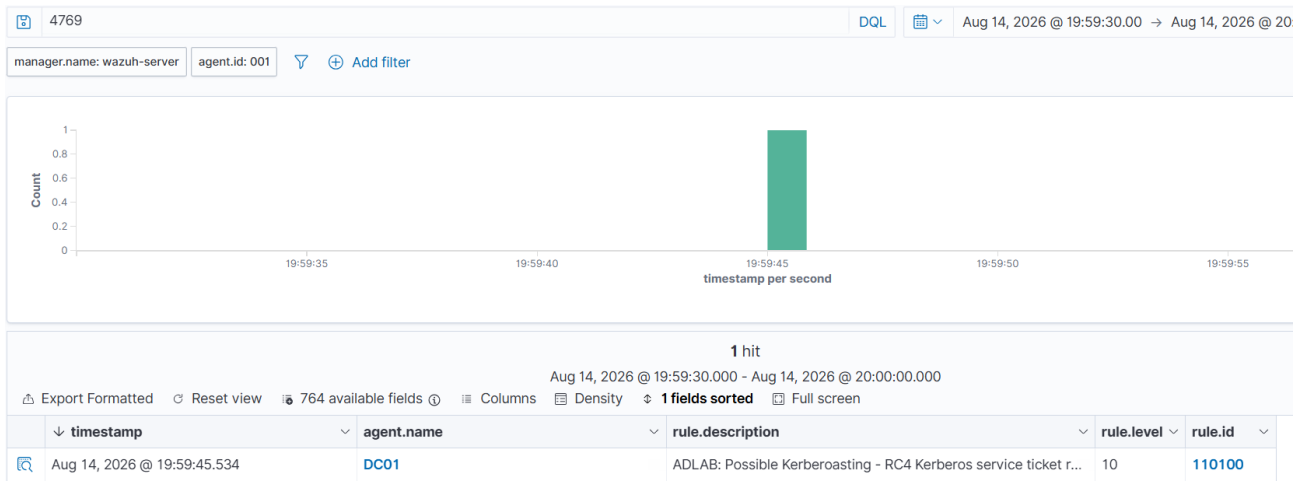
Parameter	Value
Service account	svc_web
SPN	HTTP/web.adlab.test
Primary event	4769 - Kerberos service ticket request
Encryption condition	0x17 - RC4
Custom rule	110100
Severity	Level 10
Technique	T1558.003 - Kerberoasting
Result	Detected

```
[wazuh-user@wazuh-server ~]$ sudo grep '"id":110100' /var/ossec/logs/alerts/alerts.json | tail -n 5
{"timestamp":"2026-08-14T16:59:45.534+0000","rule":{"level":10,"description":"ADLAB: Possible Kerberoasting - RC4 Kerberos service ticket requested by non-m
achine account Administrator@ADLAB.TEST","id":"110100","mitre":{"id":["T1558.003"],"tactic":["Credential Access"],"technique":["Kerberoasting"]},"firedtimes
":1,"mail":false,"groups":["adlab","windows","kerberos","kerberoasting","credential_access"],"agent":{"id":"001","name":"DC01","ip":"10.10.10.10"},"manager
":{"name":"wazuh-server"},"id":"1786726785.4124213","full_log":{"win":{"system":{"providerName":"Microsoft-Windows-Security-Auditing"},"providerGui
d":{"54849625-5478-4994-a5ba-3e3b0328c30d"},"eventID":"4769","version":"","level":"","task":"","opcode":"","keywords":"","0x8
0200000000000000"},"systemTime":"2026-08-14T16:59:15.6188453Z"},"eventRecordID":"11228","processID":"628","threadID":"2052"},"channel":{"Secu
rity"},"computer":{"DC01.adlab.test"},"severityValue":"","AUDIT_SUCCESS"},"message":["A Kerberos service ticket was requested.\\r\\n\\r\\nAccount In
formation:\\r\\n\\r\\nService Information:\\r\\n\\r\\nService Name:\\t\\tsvc_web\\r\\n\\r\\nService ID:\\t\\tS-1-5-21-36052701-3025511810-1330942903-1107\\r\\n\\r
\\nNetwork Information:\\r\\n\\r\\nClient Address:\\t\\t::ffff:10.10.10.20\\r\\n\\r\\nClient Port:\\t\\t55752\\r\\n\\r\\nAdditional Information:\\r\\n\\r\\nTicket Op
tions:\\t\\t0x40810000\\r\\n\\r\\nTicket Encryption Type:\\t\\t0x17\\r\\n\\r\\nFailure Code:\\t\\t0x0\\r\\n\\r\\nTransited Services:\\t\\t-\\r\\n\\r\\nThis event is genera
ted every time access is requested to a resource such as a computer or a Windows service. The service name indicates the resource to which access was requ
sted.\\r\\n\\r\\nThis event can be correlated with Windows logon events by comparing the Logon GUID fields in each event. The logon event occurs on the mac
hine that was accessed, which is often a different machine than the domain controller which issued the service ticket.\\r\\n\\r\\nTicket options, encryption
types, and failure codes are defined in RFC 4120.\\r\\n\\r\\n"},"eventdata":{"targetUserName":"Administrator@ADLAB.TEST","targetDomainName":"ADLAB.TEST","
serviceNames":["svc_web"],"serviceSid":"S-1-5-21-36052701-3025511810-1330942903-1107"},"ticketOptions":"","ticketEncryptionType":"","0x1
7e","ipAddress":"","::ffff:10.10.10.20","ipPort":"55752"},"status":"","0x0","logonGUID":"","{f71d186-abfb-074e-5a80-e10f371dde10}\\r\\n"},"decoder":{"na
me":"windows_eventchannel"},"data":{"win":{"system":{"providerName":"Microsoft-Windows-Security-Auditing"},"providerGuid":{"54849625-5478-4994-a5ba-3e3b0328
c30d"},"eventID":"4769","version":"","level":"","task":"","opcode":"","keywords":"","0x8020000000000000"},"systemTime":"2026-08-14T16:59:15.6188453Z"},"ev
entRecordID":"11228","processID":"628","threadID":"2052","channel":"Security","computer":"DC01.adlab.test"},"severityValue":"","AUDIT_SUCCESS"},"message":["A Ke
rberos service ticket was requested.\\r\\n\\r\\nAccount Information:\\r\\n\\r\\nAccount Name:\\t\\tAdministrator@ADLAB.TEST\\r\\n\\r\\nAccount Domain:\\t\\tADLAB.TEST\\r\\n\\r\\nLogo
n GUID:\\t\\t{f71d186-abfb-074e-5a80-e10f371dde10}\\r\\n\\r\\nService Information:\\r\\n\\r\\nService Name:\\t\\tsvc_web\\r\\n\\r\\nService ID:\\t\\tS-1-5-21-36052701-3025511810
-1330942903-1107\\r\\n\\r\\nNetwork Information:\\r\\n\\r\\nClient Address:\\t\\t::ffff:10.10.10.20\\r\\n\\r\\nClient Port:\\t\\t55752\\r\\n\\r\\nAdditional Information:\\r\\n\\r\\nTicke
t Options:\\t\\t0x40810000\\r\\n\\r\\nTicket Encryption Type:\\t\\t0x17\\r\\n\\r\\nFailure Code:\\t\\t0x0\\r\\n\\r\\nTransited Services:\\t\\t-\\r\\n\\r\\nThis event is generated every time
access is requested to a resource such as a computer or a Windows service. The service name indicates the resource to which access was requested.\\r\\n\\r\\nTh
is event can be correlated with Windows logon events by comparing the Logon GUID fields in each event. The logon event occurs on the machine that was acces
sed, which is often a different machine than the domain controller which issued the service ticket.\\r\\n\\r\\nTicket options, encryption types, and failure cod
es are defined in RFC 4120.\\r\\n\\r\\n"},"eventdata":{"targetUserName":"Administrator@ADLAB.TEST","targetDomainName":"ADLAB.TEST","serviceNames":["svc_web"],"serviceSid
":"S-1-5-21-36052701-3025511810-1330942903-1107"},"ticketOptions":"","0x40810000","ticketEncryptionType":"","0x17","ipAddress":"","::ffff:10.10.10.20","ipPort":"55752
"},"status":"","0x0","logonGUID":"","{f71d186-abfb-074e-5a80-e10f371dde10}\\r\\n"},"location":"EventChannel"}]
[wazuh-user@wazuh-server ~]$
```

Kerberos service-ticket evidence from the controlled svc_web scenario. Evidence: day3-kerberoasting-event4769-rc4.png

8.1 Wazuh rule

```
<rule id="110100" level="10">
  <if_group>authentication_success</if_group>
  <field name="win.system.eventID">^4769$</field>
  <field name="win.eventdata.ticketEncryptionType">^0x17$</field>
  <field name="win.eventdata.targetUserName" type="pcre2">^[^$]+$</field>
  <description>ADLAB: Possible Kerberoasting - RC4 Kerberos service ticket requested by non-machine account</description>
  <mitre><id>T1558.003</id></mitre>
</rule>
```



Wazuh evidence confirming the Kerberoasting detection path. Evidence: day3-wazuh-kerberoasting-live-detection.png

8.2 Sigma representation

```
detection:  
selection:  
  EventID: 4769  
  TicketEncryptionType: '0x17'  
filter_machine_accounts:  
  TargetUserName|endswith: '$'  
condition: selection and not filter_machine_accounts
```

Validation result: DETECTED

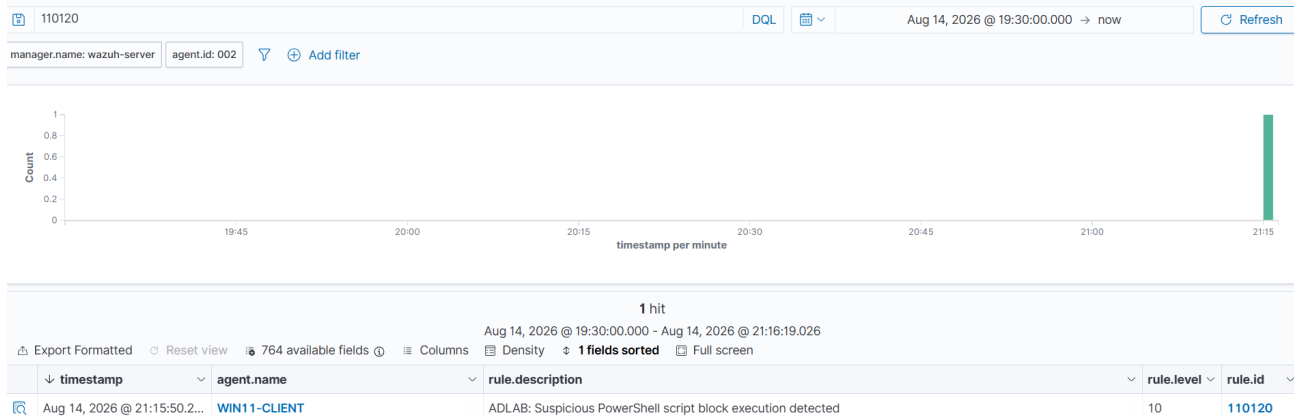
Event 4769 and the RC4 0x17 condition matched custom Wazuh rule 110100. After validation, the intentionally weak lab password used for svc_web was replaced with a stronger unique password and the account remained non-administrative.

SECTION 9

Scenario 04 - PowerShell Activity

Script Block Logging, AD discovery, and detection-content drift

Element	Current state
Primary telemetry	PowerShell Operational Event 4104; supporting Sysmon Event 1
AD discovery commands	Get-ADDomain, Get-ADUser, Get-ADGroup, Get-ADGroupMember
Sigma mapping	T1087.002 / T1069.002
Current Wazuh 110120 mapping	T1059.001 - PowerShell
Validation status	Project reports record detection, but XML/Sigma logic is not aligned



Wazuh PowerShell hunting / alert view used to review script-block activity. Evidence: day3-wazuh-suspicious-powershell-dashboard-alert.png

9.1 Sigma AD reconnaissance logic

detection:
 selection:
 ScriptBlockText | contains:
 - 'Get-ADDomain'
 - 'Get-ADUser'
 - 'Get-ADGroup'
 - 'Get-ADGroupMember'
 condition: selection

9.2 Current Wazuh 110120 logic

```
<rule id="110120" level="10">
  <if_sid>91802</if_sid>
  <field name="win.eventdata.scriptBlockText" type="pcre2">(?)i(Invoke-Expression|bIEX\b|DownloadString|Net\.WebClient|FromBase64String)</field>
  <description>ADLAB: Suspicious PowerShell script block execution detected</description>
  <mitre><id>T1059.001</id></mitre>
</rule>
```

Engineering finding

The current XML and Sigma files represent different behaviors. A production-quality repository should either align 110120 with the AD-recon Sigma rule or introduce a second rule ID for AD discovery while preserving 110120 for suspicious execution/download/encoding behavior.

Validation result: DETECTION EVIDENCE PRESENT

PowerShell 4104 telemetry and Wazuh alert/hunting evidence are present. The remaining task is semantic cleanup so the operational XML and portable Sigma rule describe the same detection intent when they are presented as equivalents.

SECTION 10

Scenario 05 - Remote Administrative Logon

Windows visibility confirmed; exact dedicated Wazuh alert not confirmed

Parameter	Value
Source	WIN11-CLIENT - 10.10.10.20
Destination	DC01 - 10.10.10.10
Method	PowerShell Remoting / WinRM
Account	ADLAB\Administrator
Observed event	4624
Logon Type	3 - Network
Authentication	Kerberos
Final state	Logged but not alerted

```
Test-NetConnection DC01 -Port 5985
$cred = Get-Credential ADLAB\Administrator
Enter-PSSession -ComputerName DC01 -Credential $cred
```

```
TimeCreated : 8/15/2026 10:02:05 AM
Id          : 4624
Message     : An account was successfully logged on.

Subject:
  Security ID:             S-1-0-0
  Account Name:            -
  Account Domain:         -
  Logon ID:                0x0

Logon Information:
  Logon Type:              3
  Restricted Admin Mode:   -
  Virtual Account:        No
  Elevated Token:         Yes

Impersonation Level:      Impersonation

New Logon:
  Security ID:             S-1-5-21-36052701-3025511810-1330942903-500
  Account Name:            Administrator
  Account Domain:         ADLAB.TEST
  Logon ID:                0x276063
  Linked Logon ID:         0x0
  Network Account Name:    -
  Network Account Domain: -
  Logon GUID:              {e4e3a2e2-7ae8-10c7-8a48-61f1e55b33cd}

Process Information:
  Process ID:              0x0
  Process Name:            -

Network Information:
  Workstation Name:        -
  Source Network Address:  10.10.10.20
  Source Port:             49927
```

Windows Security evidence for the remote administrative session. Key fields include Event ID 4624, Logon Type 3, administrative identity, and source host/IP. Evidence: test-05-remote-logon-4624.png

Validation result: LOGGED BUT NOT ALERTED

The raw Windows event was confirmed locally. An exact dedicated Wazuh alert for this specific remote administrative logon was not confirmed, so the project correctly treats this as a monitoring gap rather than a successful detection.

10.1 Recommended remediation

1. Explicitly validate Windows Security channel collection in the endpoint Wazuh configuration.
2. Create a dedicated privileged network-logon analytic using Event 4624 + Logon Type 3 + privileged identity + source context.
3. Baseline approved administration hosts and management subnets.
4. Retest with legitimate administration and a controlled unexpected source to assess false positives.

5. Add the rule to regression testing and the detection matrix.

SECTION 11

Scenario 06 - Privileged Group Membership Change

High-value Domain Admins change detection and cleanup

Parameter	Value
Test identity	helpdesk.test
Privileged group	Domain Admins
Primary event	4728
Validated Wazuh rule	60159 - Domain Admins Group Changed
Severity	Level 12
Sigma mapping	T1098 - Account Manipulation
Final result	Detected
Cleanup	helpdesk.test removed from Domain Admins

```

TimeCreated : 8/15/2026 9:56:22 AM
Id          : 4728
Message     : A member was added to a security-enabled global group.

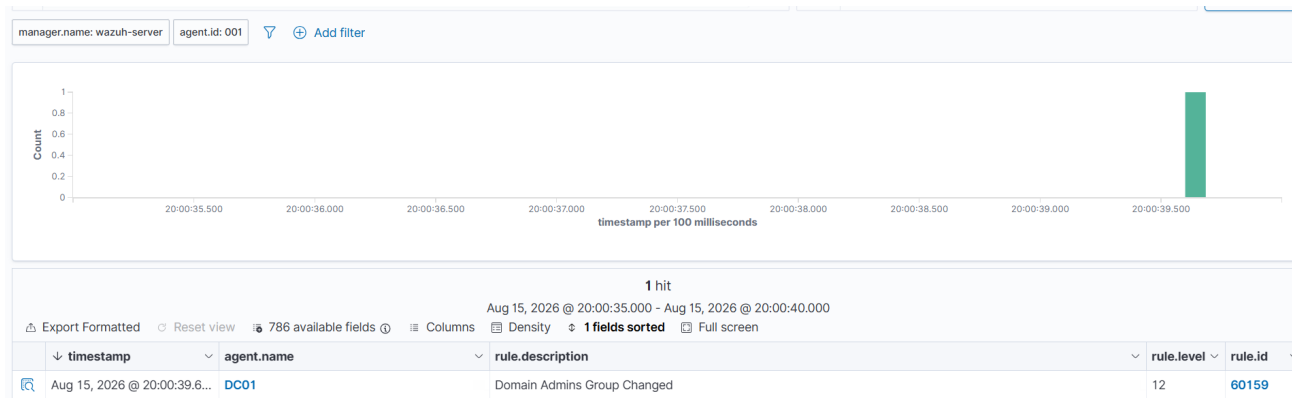
Subject:
  Security ID:      S-1-5-21-36052701-3025511810-1330942903-500
  Account Name:    Administrator
  Account Domain:  ADLAB
  Logon ID:        0x96510

Member:
  Security ID:      S-1-5-21-36052701-3025511810-1330942903-1109
  Account Name:    CN=Helpdesk Test,OU=Lab Users,DC=adlab,DC=test

Group:
  Security ID:      S-1-5-21-36052701-3025511810-1330942903-512
  Group Name:      Domain Admins
  Group Domain:    ADLAB

Additional Information:
Privileges:      -
  
```

Windows / PowerShell evidence for the controlled privileged-group membership change. Evidence: test-06-group-change-4728.png



Wazuh evidence for the privileged-group change; the validated alert is built-in rule 60159 at Level 12. Evidence: test-06-domain-admins-wazuh-rule-60159.png

11.1 Portable Sigma detection

detection:
selection_event:
EventID:

- 4728

- 4732

- 4756

selection_group:

TargetUserName | contains:

- 'Domain Admins'

- 'Enterprise Admins'

- 'Administrators'

condition: selection_event and selection_group

Validation integrity

The confirmed Wazuh evidence for this scenario is built-in rule 60159. A separate custom privilege-change Wazuh rule is not claimed as validated.

SECTION 12

Canary Deception Mechanism

High-confidence authentication tripwire using canary.admin

Property	State
Account	canary.admin
Enabled	False / disabled
Privilege	Non-privileged
Expected legitimate use	None
Observed event	4776
Source workstation	WIN11-CLIENT
Custom rule	110150
Severity	Level 12
ATT&CK context	T1078 - Valid Accounts

```
PS C:\Users\Administrator> Get-ADUser Canary.admin -Properties Enabled | Select-Object SamAccountName,Enabled
SamAccountName Enabled
-----
canary.admin      False

PS C:\Users\Administrator> Get-ADPrincipalGroupMembership Canary.admin | Select-Object Name
Name
----
Domain Users
```

Controlled canary-account test preparation and authentication attempt. Evidence: advanced-canary-attempt.png

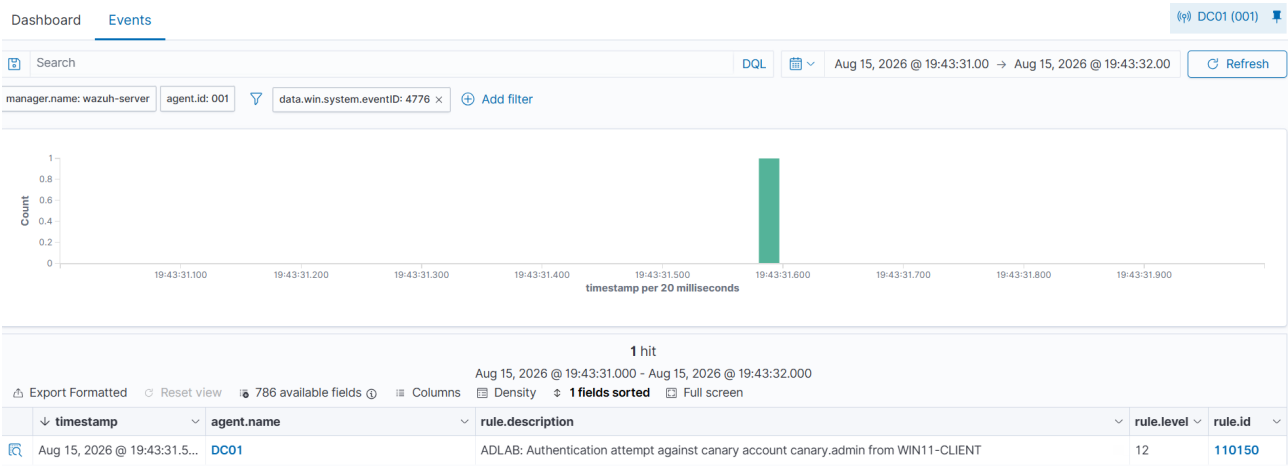
```
TimeCreated : 8/15/2026 9:34:34 AM
Id          : 4776
Message     : The computer attempted to validate the credentials for an account.

Authentication Package:  MICROSOFT_AUTHENTICATION_PACKAGE_V1_0
Logon Account:   canary.admin
Source Workstation:  WIN11-CLIENT
Error Code:      0xC000006A
```

Authentication attempt against the disabled canary identity. Evidence: advanced-canary-account-4776.png

12.1 Detection logic

```
<rule id="110150" level="12">
  <if_sid>60104</if_sid>
  <field name="win.system.eventID">^4776$</field>
  <field name="win.eventdata.targetUserName" type="pcre2">(?!^canary\.admin$</field>
  <description>ADLAB: Authentication attempt against canary account $(win.eventdata.targetUserName) from
$(win.eventdata.workstation)</description>
  <mitre><id>T1078</id></mitre>
</rule>
```



Wazuh Level-12 alert validating custom canary rule 110150. Evidence: advanced-canary-custom-rule-110150.png

Validation result: DETECTED AND RETESTED

The canary event generated Event 4776 and custom rule 110150. The detection was later revalidated after hardening and while benign background traffic was active.

SECTION 13

Targeted Active Directory Object Auditing

Event 4662 and the SACL requirement

Event 4662 is valuable for Active Directory object-access monitoring, but enabling Directory Service Access auditing alone is insufficient. The monitored object must also have an appropriate System Access Control List (SACL).

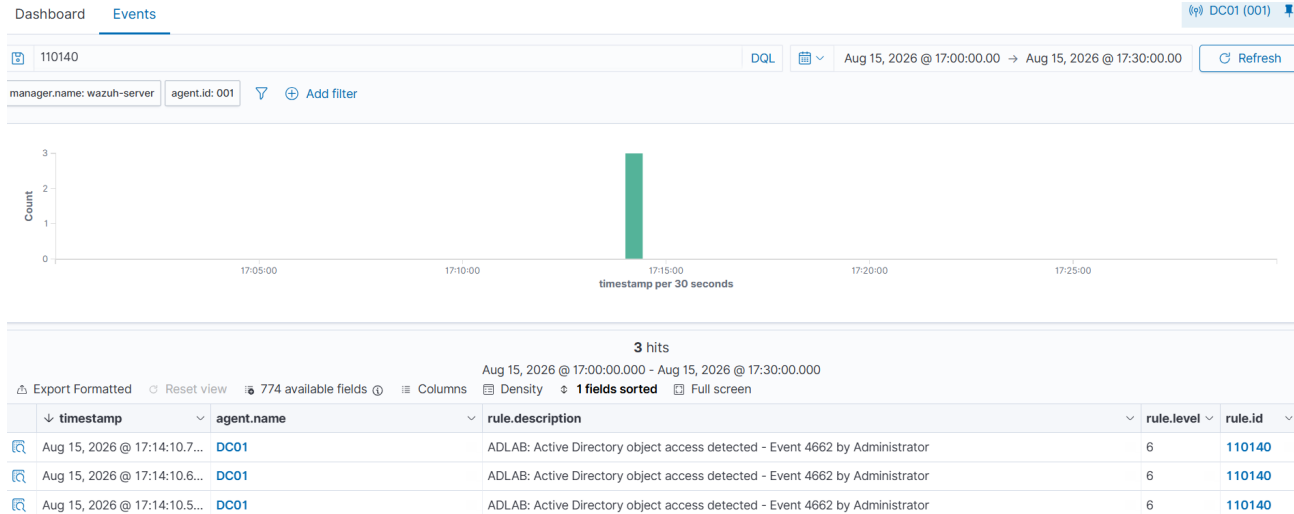
Configuration element	Lab implementation
Audit policy	Directory Service Access - Success and Failure
Object scope	OU=Test-Lab,DC=adlab,DC=test
SACL	Targeted auditing on the Test-Lab OU
Validation action	Harmless OU description change
Expected telemetry	Event 4662
Result	Generated locally and searchable in Wazuh

```
PS C:\Users\Administrator> gpupdate /force
Updating policy...

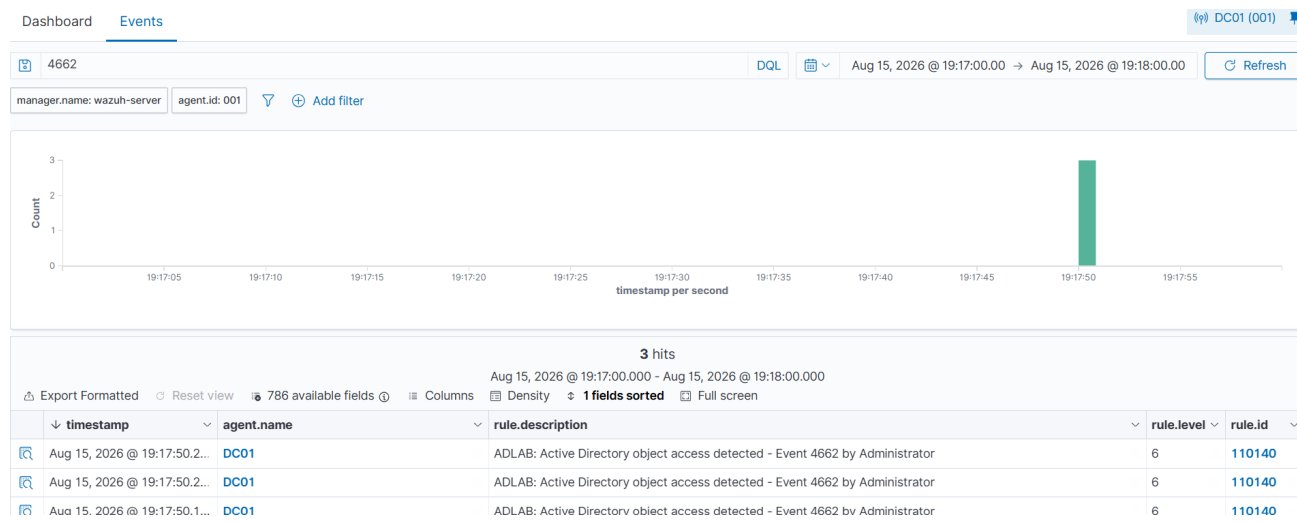
Computer Policy update has completed successfully.
User Policy update has completed successfully.

PS C:\Users\Administrator> auditpol /get /subcategory:"Directory Service Access"
System audit policy
Category/Subcategory          Setting
DS Access
  Directory Service Access    Success and Failure
PS C:\Users\Administrator> auditpol /get /subcategory:"Directory Service Changes"
System audit policy
Category/Subcategory          Setting
DS Access
  Directory Service Changes    Success
PS C:\Users\Administrator> █
```

Directory Service Access policy verification used before Event 4662 testing. Evidence: day3-directory-service-audit-policy.png



Wazuh-side AD object-access evidence from the 4662 validation workflow. Evidence: day3-wazuh-directory-service-event4662.png



Additional Wazuh evidence for targeted Active Directory object activity. Evidence: day3-wazuh-hunting-ad-object-activity.png

Set-ADOrganizationalUnit `

-Identity "OU=Test-Lab,DC=adlab,DC=test" `

-Description "4662 audit validation"

Historical evidence note

Earlier lab evidence included a Wazuh 4662 custom-rule view using rule ID 110140. The current repository local_rules.xml does not contain 110140, so this report does not present 110140 as a current active rule. The current supported claim is that Event 4662 was generated and searchable in Wazuh.

Validation result: TELEMETRY VALIDATED

The SACL + audit-policy combination successfully generated targeted object-access telemetry.

SECTION 14

LDAP Diagnostic Visibility

Temporary NTDS Event 1644 testing and central-ingestion gap

NTDS Field Engineering diagnostics were enabled temporarily to generate Event 1644 and assess LDAP query visibility. This was a diagnostic test, not a permanent logging configuration.

```
TimeCreated : 8/15/2026 9:44:35 AM
Id          : 1644
Message     : Internal event: A client issued a search operation with the following options.

Client:
[::1]:49859
Starting node:
CN=Schema,CN=Configuration,DC=adlab,DC=test
Filter:
( & (objectClass=attributeSchema) (LDAPDisplayName=msds-allowedtoactonbehalffotheridentity) )
Search scope:
subtree
Attribute selection:
attributeSyntax,oMSyntax,oMObjectClass
Server controls:

Visited entries:
1
Returned entries:
1
Used indexes:
idx_LDAPDisplayName:1:N;
Pages referenced:
39
Pages read from disk:
1
Pages pre-read from disk:
0
Clean pages modified:
0
Dirty pages modified:
0
Search time (ms):
15
Attributes Preventing Optimization:
none
User:
ADLAB\DC01$
```

Local DC01 Event 1644 showing LDAP search diagnostic details, including search scope, filter, indexes, pages referenced, and search time.
Evidence: day3-ldap-1644.png

14.1 Temporary diagnostic configuration

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\NTDS\Diagnostics" /v "15 Field Engineering" /t REG_DWORD /d 5 /f
reg add "HKLM\SYSTEM\CurrentControlSet\Services\NTDS\Parameters" /v "Search Time Threshold (msecs)" /t REG_DWORD /d 1 /f
```

14.2 Cleanup

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\NTDS\Diagnostics" /v "15 Field Engineering" /t REG_DWORD /d 0 /f
reg delete "HKLM\SYSTEM\CurrentControlSet\Services\NTDS\Parameters" /v "Search Time Threshold (msecs)" /f
```

Validation result: LOCAL TELEMETRY ONLY

Event 1644 was successfully generated locally on DC01. Wazuh ingestion was not confirmed. The final project therefore treats this as a telemetry gap, not a successful SIEM detection.

14.3 Recommended remediation

- Explicitly configure Directory Service channel collection where required.
- Generate a controlled LDAP query and verify event transport end-to-end.
- Confirm Wazuh decoding and searchable fields before building alert logic.
- Keep verbose Field Engineering logging disabled outside short diagnostic windows.

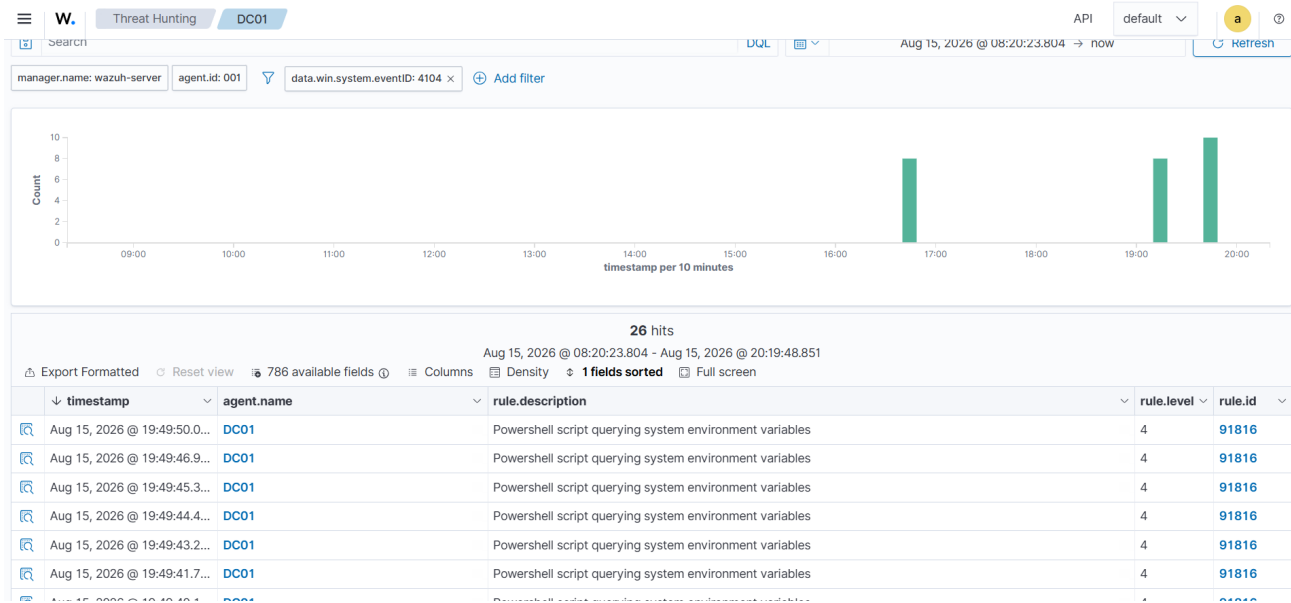
SECTION 15

Threat Hunting Workflows

Repeatable analyst searches for investigation and validation

The repository includes repeatable Wazuh searches in dashboards/wazuh-saved-searches.md. These hunts turn one-time lab tests into reusable analyst workflows.

Hunt	Example query / focus
Authentication	data.win.system.eventID:(4624 OR 4625 OR 4771 OR 4776)
Kerberos	data.win.system.eventID:4769 / rule.id:110100
PowerShell	data.win.system.eventID:4104 / rule.id:110120
Privilege changes	data.win.system.eventID:(4728 OR 4732 OR 4756) / rule.id:60159
AD object activity	data.win.system.eventID:(4662 OR 5136)
Canary	rule.id:110150
Network reconnaissance	rule.id:110130
LDAP diagnostic	data.win.system.eventID:1644 - documented, but Wazuh ingestion not validated



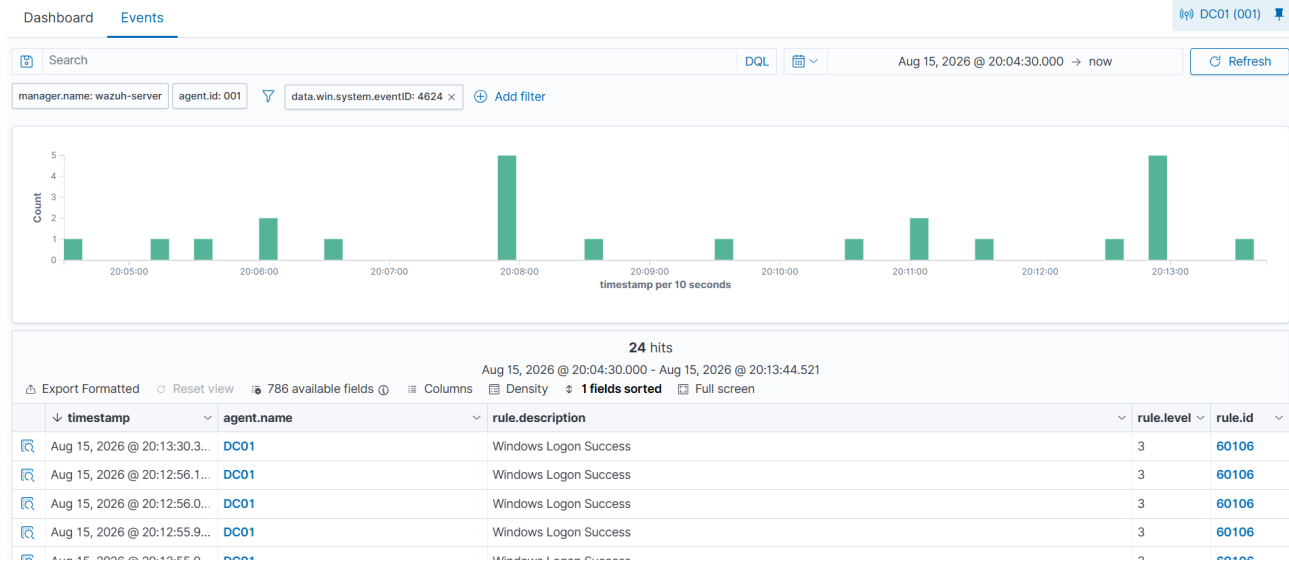
Representative Wazuh hunting view used to investigate event distributions and alert context. Evidence: day3-wazuh-hunting-authentication.png

Document Details

[View surrounding documents](#)[View single document](#)

data.win.eventdata.targetUserName	DC01\$@ADLAB.TEST
data.win.eventdata.ticketEncryptionType	0x12
data.win.eventdata.ticketOptions	0x40800000
data.win.system.channel	Security
data.win.system.computer	DC01.adlab.test
data.win.system.eventID	4769
data.win.system.eventRecordID	14642
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A Kerberos service ticket was requested. Account Information: Account Name: DC01\$@ADLAB.TEST Account Domain: ADLAB.TEST Logon GUID: {221e402d-1f72-8edd-6d76-f2308e16451a}
data.win.system.opcode	0
data.win.system.processID	640

Expanded event detail view used to inspect decoded fields during investigation. Evidence: advanced-canary-account-event-details.png



Additional Wazuh hunting view demonstrating time-based event analysis. Evidence: day3-wazuh-hunting-kerberos.png

15.1 Analyst pivot fields

- agent.name and computer
- event ID and event provider
- target and subject usernames
- source IP and workstation
- logon type and authentication package
- Kerberos encryption type and service name
- PowerShell scriptBlockText
- Wazuh rule ID, description, and level
- timestamps and adjacent activity

SECTION 16

Benign Noise and False-Positive Validation

Testing detections in a non-silent lab

A detection tested only in a silent lab can appear more precise than it would under normal endpoint activity. The project therefore introduced low-rate enterprise-like background traffic while retesting selected custom rules.

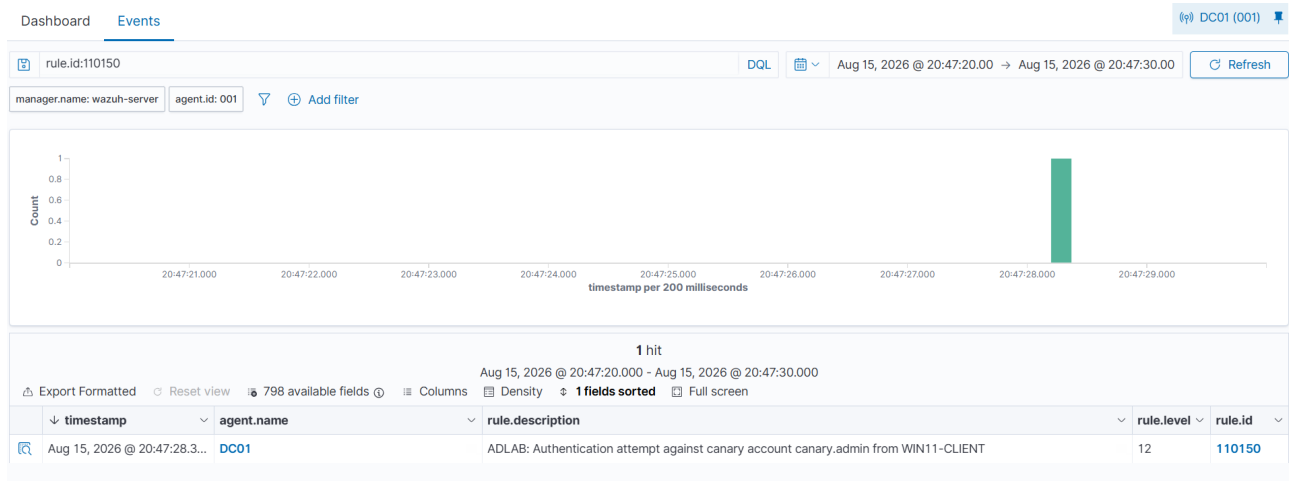
Benign activity	Purpose
DNS lookups	Routine name-resolution traffic
ICMP checks	Low-rate connectivity activity
SYSVOL reads	Normal domain resource access
NETLOGON reads	Normal domain resource access
PowerShell activity	Routine administrative / scripting noise

```

273 WIN11-CLIENT unknown unknown
260 WIN11-CLIENT Microsoft-Windows-Sysmon/Operational 11
237 DC01 unknown unknown
180 DC01 Microsoft-Windows-Sysmon/Operational 11
131 DC01 Microsoft-Windows-Sysmon/Operational 3
49 DC01 Security 4624
47 DC01 Microsoft-Windows-Sysmon/Operational 22
46 DC01 Security 4672
42 WIN11-CLIENT Microsoft-Windows-Sysmon/Operational 1
35 WIN11-CLIENT Microsoft-Windows-Sysmon/Operational 5
30 DC01 System 7036
27 WIN11-CLIENT Microsoft-Windows-Sysmon/Operational 3
21 DC01 Security 5379
14 WIN11-CLIENT Security 4798
11 DC01 Microsoft-Windows-Sysmon/Operational 5
10 WIN11-CLIENT System 10010
10 DC01 Security 4688
10 DC01 Microsoft-Windows-Sysmon/Operational 1
9 WIN11-CLIENT Security 5379
7 WIN11-CLIENT Security 4672
7 WIN11-CLIENT Security 4624
7 WIN11-CLIENT Microsoft-Windows-PowerShell/Operational 4104
6 DC01 Security 4769
5 WIN11-CLIENT Microsoft-Windows-Sysmon/Operational 22
2 WIN11-CLIENT System 566
2 WIN11-CLIENT Security 4648
2 WIN11-CLIENT Security 4634
2 WIN11-CLIENT Application 15
2 DC01 Security 4768
1 WIN11-CLIENT Application 0

```

Benign baseline event distribution captured during the background-noise test. Evidence: day4-benign-baseline-event-distribution.png



Wazuh evidence used during false-positive validation with background traffic present. Evidence: day4-wazuh-network-scan-benign-false-positive.png

Detection	Controlled activity detected	Observed benign FP
110130 - network reconnaissance	Yes	None observed during validation

Detection	Controlled activity detected	Observed benign FP
		window
110150 - canary authentication	Yes	None observed during validation window

Correct conclusion

No benign false positive was observed during the controlled validation window. This does not prove a zero false-positive rate in a production enterprise.

Repository reproducibility gap

The current main branch contains zero-byte automation/benign-noise.ps1 and automation/scheduled-task-setup.ps1 files. The reports and screenshots show that the workflow was executed, but the executable source should be restored for reproducible regression testing.

SECTION 17

Hardening and Post-Hardening Retest

Removing test weaknesses without breaking visibility

Control	Final state
svc_web password	Intentionally weak lab password replaced with stronger unique password
svc_web privilege	Non-administrative; Domain Users only
helpdesk.test	Removed from Domain Admins
canary.admin	Disabled and non-privileged
LDAP Field Engineering	Returned to 0 / verbose diagnostics disabled
Temporary LDAP threshold	Removed
Sysmon	Monitoring retained
Wazuh agents	Monitoring retained

```

PS C:\Users\Administrator> Get-ADPrincipalGroupMembership svc_web | Select-Object Name

Name
----
Domain Users

PS C:\Users\Administrator> Get-ADGroupMember "Domain Admins" |
>> Select-Object Name,SamAccountName

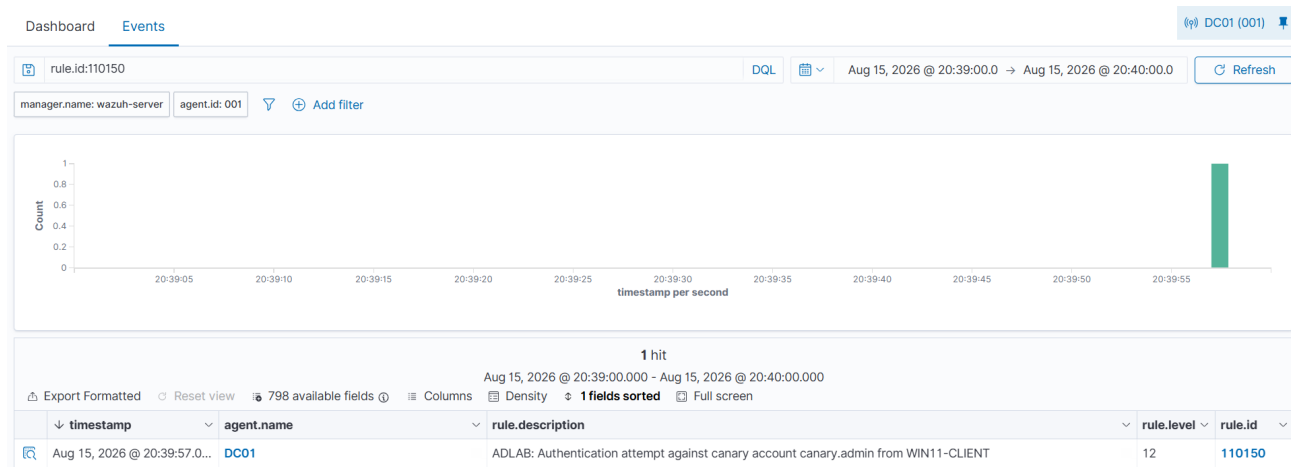
Name          SamAccountName
----          -
Administrator Administrator

PS C:\Users\Administrator> Get-ADUser canary.admin -Properties Enabled |
>> Select-Object SamAccountName,Enabled

SamAccountName Enabled
-----
canary.admin    False

```

Post-hardening identity and account-state validation. Evidence: day7-hardening-account-state.png



Post-hardening Wazuh evidence showing that the detection pipeline remained operational. Evidence: day7-retest-canary-detection.png

Validation result: REMEDIATION + RETEST COMPLETE

The canary detection was successfully triggered after hardening, demonstrating that security improvements did not remove the monitoring path.

SECTION 18

Validation Matrix and Engineering Findings

Final detection status and interpretation

ID	Scenario	Telemetry	Detection / outcome	Status
ADLAB-001	Password spray	4625 / 4771 / 4776	110110	Detected
ADLAB-002	Kerberoasting	4769	110100	Detected
ADLAB-003	PowerShell	4104 / Sysmon 1	110120 documented; content drift noted	Detected in validation
ADLAB-004	Privileged group change	4728	60159 Level 12	Detected
ADLAB-005	Canary authentication	4776	110150 Level 12	Detected
ADLAB-006	Remote admin logon	4624 Type 3	Exact dedicated alert not confirmed	Logged, not alerted
ADLAB-007	AD object access	4662	Searchable in Wazuh	Telemetry validated
ADLAB-008	LDAP diagnostic	1644	Wazuh ingestion not confirmed	Telemetry gap

18.1 High-value findings

- Correlation meaningfully improved network-scan and password-spray detections compared with one-event alerting.
- A dedicated canary identity produced a high-confidence authentication signal with little expected legitimate activity.
- Privileged-group changes generated high-value Windows telemetry and a Level-12 Wazuh alert.
- Event 4662 monitoring required both audit policy and object-level SACL configuration.
- Remote administrative activity was visible locally but did not have a confirmed dedicated Wazuh alert.
- LDAP Event 1644 remained a central-ingestion gap.
- PowerShell detection-as-code needs semantic alignment between Wazuh and Sigma.
- Detection latency was not measured consistently and should be added to future test methodology.

SECTION 19

Limitations and Engineering Recommendations

Technical debt, remediation priorities, and next-step maturity

Priority	Finding	Recommended action
P1	Remote Type-3 administrative logon not alerted	Build dedicated 4624 Type 3 privileged-logon analytic; baseline approved source hosts; regression test.
P1	Windows Security collection not explicit in repo agent.conf	Document or explicitly configure the Security channel used by the lab.
P2	LDAP 1644 not confirmed in Wazuh	Validate Directory Service transport, decoding, and searchable fields before detection engineering.
P2	PowerShell Wazuh/Sigma drift	Separate suspicious PowerShell execution from AD discovery or align the rule implementations.
P2	Benign-noise scripts are empty on main	Restore source and task-setup script; document execution and cleanup.
P3	No measured alert latency	Record T0 activity, T1 endpoint event, T2 ingestion, T3 alert timestamps.
P3	Sigma validation manual only	Add Sigma schema/CLI validation in CI.
P3	No automated Wazuh regression suite	Create event fixtures and automated rule tests.

19.1 Recommended next maturity phase

- Close telemetry gaps before adding many more attack techniques.
- Treat every detection as code with owner, ATT&CK mapping, severity, expected fields, tuning notes, and regression tests.
- Measure event-to-alert latency and track false-positive observations across repeated runs.
- Add a second Domain Controller or Windows Event Forwarding layer to test distributed telemetry.
- Translate Sigma detections to a second SIEM backend to demonstrate portability.
- Add identity-focused or EDR telemetry only after the current pipeline is fully reproducible.

Engineering priority

The project does not need more attack volume to become stronger. Its biggest gains now come from reproducibility, measurement, rule alignment, and closing the two documented visibility gaps.

SECTION 20

Appendices

Rule reference, configuration inventory, repository artifacts, and evidence catalog

Appendix A - Full custom Wazuh rule inventory

```
<group name="adlab, windows, kerberos,">
  <rule id="110100" level="10">
    <if_group>authentication_success</if_group>
    <field name="win.system.eventID">^4769$</field>
    <field name="win.eventdata.ticketEncryptionType">^0x17$</field>
    <field name="win.eventdata.targetUserName" type="pcre2">^[^$]+</field>
    <description>ADLAB: Possible Kerberoasting - RC4 Kerberos service ticket requested by non-machine account $(win.eventdata.targetUserName)</description>
    <mitre><id>T1558.003</id></mitre>
    <group>kerberoasting, credential_access,</group>
  </rule>
</group>

<group name="adlab, windows, authentication,">
  <rule id="110110" level="12" frequency="4" timeframe="120">
    <if_matched_sid>60122</if_matched_sid>
    <same_field>win.eventdata.ipAddress</same_field>
    <different_field>win.eventdata.targetUserName</different_field>
    <description>ADLAB: Possible password spraying - multiple user accounts failed authentication from the same source IP</description>
    <mitre><id>T1110.003</id></mitre>
    <group>password_spray, authentication_failed, credential_access,</group>
  </rule>
</group>

<group name="adlab, powershell, windows,">
  <rule id="110120" level="10">
    <if_sid>91802</if_sid>
    <field name="win.eventdata.scriptBlockText" type="pcre2">(?!)(Invoke-Expression|&#160;bIEX\b|DownloadString|Net.WebClient|FromBase64String)</field>
    <description>ADLAB: Suspicious PowerShell script block execution detected</description>
    <mitre><id>T1059.001</id></mitre>
    <group>powershell, suspicious_script, execution,</group>
  </rule>
</group>

<group name="adlab, windows, sysmon, reconnaissance,">
  <rule id="110129" level="1">
    <if_group>sysmon_event3</if_group>
    <field name="win.system.eventID">^3$</field>
    <field name="win.eventdata.initiated">^false$</field>
    <description>ADLAB: Sysmon inbound connection candidate for network scan correlation</description>
    <options>no_log</options>
    <group>network_scan_candidate,</group>
  </rule>

  <rule id="110130" level="10" frequency="5" timeframe="10">
    <if_matched_sid>110129</if_matched_sid>
    <same_field>win.eventdata.sourceIp</same_field>
    <same_field>win.eventdata.destinationIp</same_field>
    <different_field>win.eventdata.destinationPort</different_field>
    <description>ADLAB: Possible network service scanning from $(win.eventdata.sourceIp) against $(win.eventdata.destinationIp)</description>
    <mitre><id>T1046</id></mitre>
    <group>network_service_discovery, reconnaissance, discovery,</group>
  </rule>
</group>

<group name="windows, authentication, adlab, canary,">
  <rule id="110150" level="12">
    <if_sid>60104</if_sid>
    <field name="win.system.eventID">^4776$</field>
    <field name="win.eventdata.targetUserName" type="pcre2">(?!)(?i)^canary\.admin$</field>
    <description>ADLAB: Authentication attempt against canary account $(win.eventdata.targetUserName) from $(win.eventdata.workstation)</description>
    <mitre><id>T1078</id></mitre>
  </rule>
</group>
```

Appendix B - Sigma rule inventory

File	Purpose	Level	ATT&CK
kerberoasting.yml	RC4 Kerberos service-ticket	high	T1558.003

File	Purpose	Level	ATT&CK
	behavior for non-machine accounts		
powershell-ad-recon.yml	Get-AD* domain account/group discovery	medium	T1087.002 / T1069.002
privileged-group-change.yml	User added to privileged AD / Windows security group	high	T1098

Appendix C - Repository structure

```

active-directory-attack-detection-lab/
|-- automation/
|   |-- benign-noise.ps1           [currently empty on main]
|   |-- create-lab-users.ps1
|   |-- scheduled-task-setup.ps1   [currently empty on main]
|-- configs/
|   |-- sysmon/sysmon-config.xml
|   |-- windows-auditing/advanced-audit-policy.md
|   |-- windows-auditing/powershell-logging.md
|   |-- wazuh/agent.conf
|-- detections/
|   |-- wazuh/local_rules.xml
|   |-- sigma/
|       |-- kerberoasting.yml
|       |-- powershell-ad-recon.yml
|       |-- privileged-group-change.yml
|-- dashboards/wazuh-saved-searches.md
|-- diagrams/
|   |-- architecture.png
|   |-- architecture-detailed.png
|-- reports/
|   |-- detection-validation-matrix.md
|   |-- final-summary.md
|   |-- noise-baseline.md
|   |-- test-01-reconnaissance.md
|   |-- test-02-password-spray.md
|   |-- test-03-kerberoasting.md
|   |-- test-04-powershell.md
|   |-- test-05-lateral-movement.md
|   |-- test-06-privilege-change.md
|-- screenshots/
`-- docs/

```

Appendix D - Evidence catalog

Evidence file	Evidence file	Evidence file
advanced-canary-account-4776.png	advanced-canary-account-event-details.png	advanced-canary-account-wazuh-4776.png
advanced-canary-attempt.png	advanced-canary-custom-rule-110150.png	canary-account-disabled.png
day1-ad-ds-installed.png	day1-dc01-ip-hostname.png	day1-domain-controller.png
day1-virtualbox-ad-lab-network.png	day2-domain-join.png	day2-domain-user-login.png
day2-ou-structure.png	day2-svc-web-spn.png	day2-workstation-ou.png
day3-advanced-ad-auditing-4662.png	day3-audit-policy-1.png	day3-audit-policy-2.png
day3-dc01-gpo-result.png	day3-directory-service-audit-policy.png	day3-directory-service-event4662.png
day3-encoded-powershell-test-command.png	day3-kerberoasting-event4769-rc4.png	day3-kerberos-service-ticket.png
day3-ldap-1644.png	day3-powershell-4104-event.png	day3-powershell-test-command.png
day3-sysmon-dc01-config-validation.png	day3-sysmon-dc01-event1.png	day3-sysmon-win11-dns-event22.png
day3-sysmon-win11-event1.png	day3-sysmon-win11-running.png	day3-wazuh-adlab-connectivity.png
day3-wazuh-agents-status.png	day3-wazuh-all-agents-active.png	day3-wazuh-dashboard.png
day3-wazuh-dc01-agent-active.png	day3-wazuh-dc01-agent-deployment.png	day3-wazuh-dc01-agent-running.png
day3-wazuh-directory-service-event4662.png	day3-wazuh-encoded-powershell-dashboard-alert.png	day3-wazuh-failed-logon-detection.png
day3-wazuh-hunting-ad-object-activity.png	day3-wazuh-hunting-authentication.png	day3-wazuh-hunting-kerberos.png
day3-wazuh-hunting-powershell.png	day3-wazuh-hunting-privilege-changes.png	day3-wazuh-kerberoasting-dashboard-alert.png
day3-wazuh-kerberoasting-live-detection.png	day3-wazuh-kerberoasting-rule-logtest.png	day3-wazuh-networking.png
day3-wazuh-password-spray-dashboard-alert.png	day3-wazuh-suspicious-powershell-dashboard-alert.png	day3-win11-powershell-4104.png
day3-win11-workstation-logging-gpo.png	day4-benign-baseline-event-distribution.png	day4-benign-baseline-script.png
day4-benign-noise-automated-run.png	day4-benign-noise-scheduled-task.png	day4-dc01-kali-multiport-recon.png
day4-dc01-sysmon-kali-recon.png	day4-kali-dc01-connectivity.png	day4-kali-dc01-service-recon.png
day4-kali-dns-route.png	day4-wazuh-kali-recon-telemetry.png	day4-wazuh-network-scan-benign-false-positive.png
day4-wazuh-network-scan-false-positive-check.png	day4-wazuh-network-scan-tuned-detection.png	day4-wazuh-network-service-scan-dashboard-alert.png
day7-canary-noise-validation.png	day7-hardening-account-state.png	day7-retest-canary-detection.png

Evidence file	Evidence file	Evidence file
test-05-remote-logon-4624.png	test-06-domain-admins-wazuh-rule-60159.png	test-06-group-change-4728.png
test-06-group-change-wazuh.png	wazuh-logtest-validation-tool.png	

Appendix E - Source files reviewed

- detections/wazuh/local_rules.xml
- detections/sigma/kerberoasting.yml
- detections/sigma/powershell-ad-recon.yml
- detections/sigma/privileged-group-change.yml
- configs/sysmon/sysmon-config.xml
- configs/wazuh/agent.conf
- configs/windows-auditing/advanced-audit-policy.md
- configs/windows-auditing/powershell-logging.md
- automation/create-lab-users.ps1
- dashboards/wazuh-saved-searches.md
- reports/detection-validation-matrix.md
- reports/noise-baseline.md
- reports/test-01 through test-06
- docs/FINAL.md
- repository screenshots and architecture diagrams

Final assessment

This project is strongest when presented as a detection-engineering case study rather than an attack lab: the evidence shows telemetry design, correlation, validation, hunting, noise testing, remediation, and explicit gap analysis.

END OF TECHNICAL ENGINEERING REPORT

All activities documented here were performed in the authorized isolated AD-LAB environment.